DATE 16/09/2022 PAGE

DEEP LEARNING

ASSIGNMENT-1

Ans. (c)

SUBMITTED TO:

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Date of submission:

16/09/2022

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Implement XOR function using MP Neuron

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

XOR function is represent as $y = x_1\overline{x_2} + \overline{x_1}x_2$

$$y = x_1\overline{x_2} + \overline{x_1}x_2$$

$$= z_1 + z_2$$

or

$$= z_1 \cup z_2$$

Compute the net inputs

$w_1 = 1$, $w_2 = -1$

(0,0) z_1 in = 0x1 + 0x-1 = 0

(0,1) z_1 in = 0x1 + 1x-1 = -1

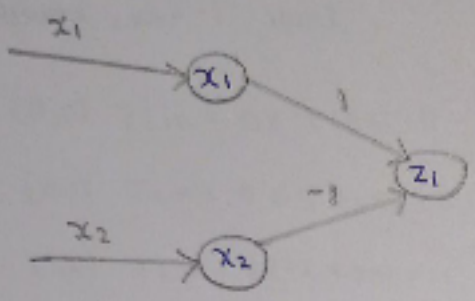
(1,0) z_1 in = 1x1 + 0x-1 = 1

(1,1) z_1 in = 1x1 + 1x-1 = 0

On the basis of this input this is possible to get the required output as $w_{11} = 1$ and $w_{21} = -1$.

Condition $\theta \geq 1$ for the z_1 neuron

$z_1 \cdot x_1\overline{x_2}$



$z_1 = x_1 \bar{x_2}$

	x_1	$\bar{x_2}$	z_1
(0,0)	0	1	0
(0,1)	1	1	1
(1,0)	0	0	0
(1,1)	1	0	0

$z_2 = \bar{x_1} x_2$

	$\bar{x_1}$	x_2	z_1
(0,0)	1	1	0
(1,0)	0	1	1
(1,1)	0	0	0
(0,1)	1	0	1

Compute net input Z_2

Assume, $w_1 = -1$ $w_2 = 1$

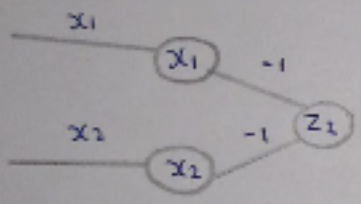
(0,0) $Z_2in = 0 \times -1 + 0 \times 1 = 0$

(0,1) $Z_2in = 0 \times -1 + 1 \times 0 = 1$

(1,0) $Z_2in = 1 \times -1 + 0 \times 1 = -1$

(1,1) $Z_2in = 1 \times -1 + 1 \times 1 = 0$

On the basis of the net input it is possible to get requested output. Hence $w_{21} = -1$, $w_{22} = 1$



The final neural network

